What Lifestyle and Behavioral Habits Predict BMI in Young People A Regression Analysis Using the Young People Survey

Student name: Khanzada Mubashir Hassan

Course: Introduction to Python

Email: mkhanzada@edu.hse.ru

March 16, 2026

Contents

1 Research Question 2

2 Data and Variables 3

3 Exploratory Data Analysis 3

3.1 BMI Distribution 4

3.2 Lifestyle Predictors 4

3.3 Correlations 5

3.4 Gender Differences 7

3.5 Scatterplots 8

4 Regression Methods 8

4.1 Ordinary Least Squares 8

4.2 Ridge Regression 9

5 Results 10

5.1 OLS Regression 10

5.2 Residual Diagnostics 12

5.3 Ridge Regression 13

6 Model Comparison 13

7 Limitations 14

8 Conclusion 16

1 Research Question

In this report I want to answer the question which is that whether it is possible to predict the body mass index (BMI) from the behavior habits and self-reported life style in the young people. BMI is widely used as a health indicator. We can calculate it from height and weight as: BMI = w eight (kg)/ (height (cm)/100)2. In vernal BMI should not be a perfect measure of the health but it is very to compute ad the younger people's dataset contains the both height and weight so I calculate It directly from there. In this I am very interested in behavioral variables: what is peoples eating's. how often peoples eat. How much they in day-to-day life they drink

and smoke and also some of their personality related things like workaholism and self-Criticism. I also add the age and gender as a demographic control, because these can be certainly affecting BMI regardless of the lifestyle. In this the Particular research question is: which self-reported lifestyle habits are correlated with BMI in young people, and how skillfully can we predict BMI from these habits?

To answer this question. I used two regression models Ridge regression and OLS linear regression. I decided to go with regression because the BMI is a continuous numeric variable, it is not a class or category. Ridge and OLS to find out whether regularization help out when predictors might be correlated with each other.

2 Data and Variables

The young people's survey in mostly conducted on the students in Slovakia. The full data set contain 1010 rows and 150 columns. I used the height and weight column to compute the BMI from the data set. Then I used filter to take out the rows where the BMI is less then 14 and above 50, because those values most likely is the data entry errors for example, one person had computed the BMI greater than 140, which Is purely impossible. This filter leaves 979 rows after removing 31 from it. Afterwards, I chose 13 predictor variables and remove any rows that contain missing values. The final modeling sample is n = 924.

I targeted the BMI variable. So, it has the mean of 21.83 and it contain the stander deviation of 23.07 for this sample. The maximum value in this is 39.2 and and the minimum is 15.5 and the median is 21.45 which is slightly lower than the mean, and it is already suggested in right skew.

I choose 13 predictors from multum areas of the life. in which the three are categorical and I start encode then as ordered numbers. I do this like for smoking: 0 who never smoked, 1 = who tried smoking, 2 = ex-smoker, 3 = who is currently smoking, same for the Alcohol: 0 = who never, 1 = a social drinker, 2 = too much drink. For Gender: 0 = female, 1 = male. The other remaining 10 variables are the numeric Likert scales from 1 to 5. To provide some context for the data: Healthyeatnghasmean3.04 (SD0.94), Actvesport3.28(SD1.51), Passive sport 3.39 (SD 1.41), Spending on gadgets 2.88 (SD 1.29), and Workaholism has mean exactly 3.00 (SD 1.28), and it is precisely in the middle of the scale. The age ranges from 15 to 30 and have a mean of 20.4 years.

I mainly include healthy eating and Active sport because they should obviously be connected to body weight. I use gadget spending as a crude indicator of sedentary behavior. I use personality variables like self-criticism and workaholism to see if they make a difference in addition to lifestyle variables. To be honest, I didn't think they would be substantial, but I wanted to give it a try.

3 Exploratory Data Analysis

3.1 BMI Distribution

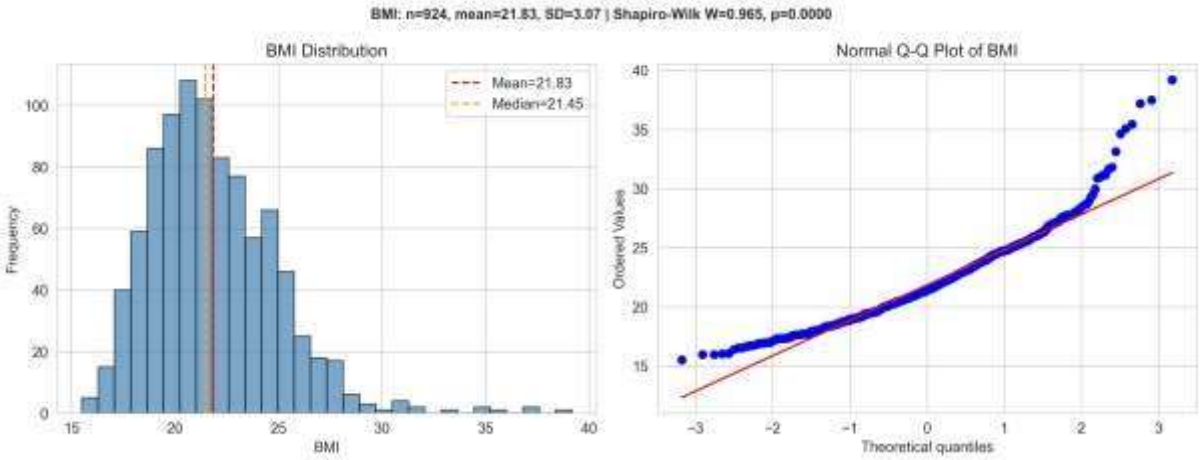


Figure 1: BMI distribution (left) and normal Q-Q plot (right). Red dashed line shows the mean (21.83), orange dashed line the median (21.45).

as seen in Figure 1 the BMI is right skewed in this sample. Kurtosis is 2.96 and skewness is 1.10. The distribution is not such a normal, according to the Shapiro-Wilk test, that yields W = 0.9646 with $p < 0.001$. The Q-Q graphic shows that the right tail lifts over the diagonal starting at about BMI 28. This skew is actually expected. Approximately respondents are lean and young, so most of the value's cluster is less then 25. A shot group which has higher BMI values is starched the distribution the right. The non-normality is not directly invalidating OLS regression the thing maters for inference is the normality of residuals, not the outcome variable itself. But we have this thing in mind when we are interpreting the results.

3.1 Lifestyle Predictors

Figure 2 shows the distributions of six Likert-scale predictors. The Active sport containing the widest spread (SD = 1.51), it means respondents are vary quite a lot in how they are physically active. Energy levels (mean 3.63) and Happiness in life (mean 3.71) both are slightly above the middle of the scale. Workaholism has a mean of exactly 3.00 the very center of the 1-5 range

so on average respondents are neither particularly nor lazy on average particularly hard-working on average.

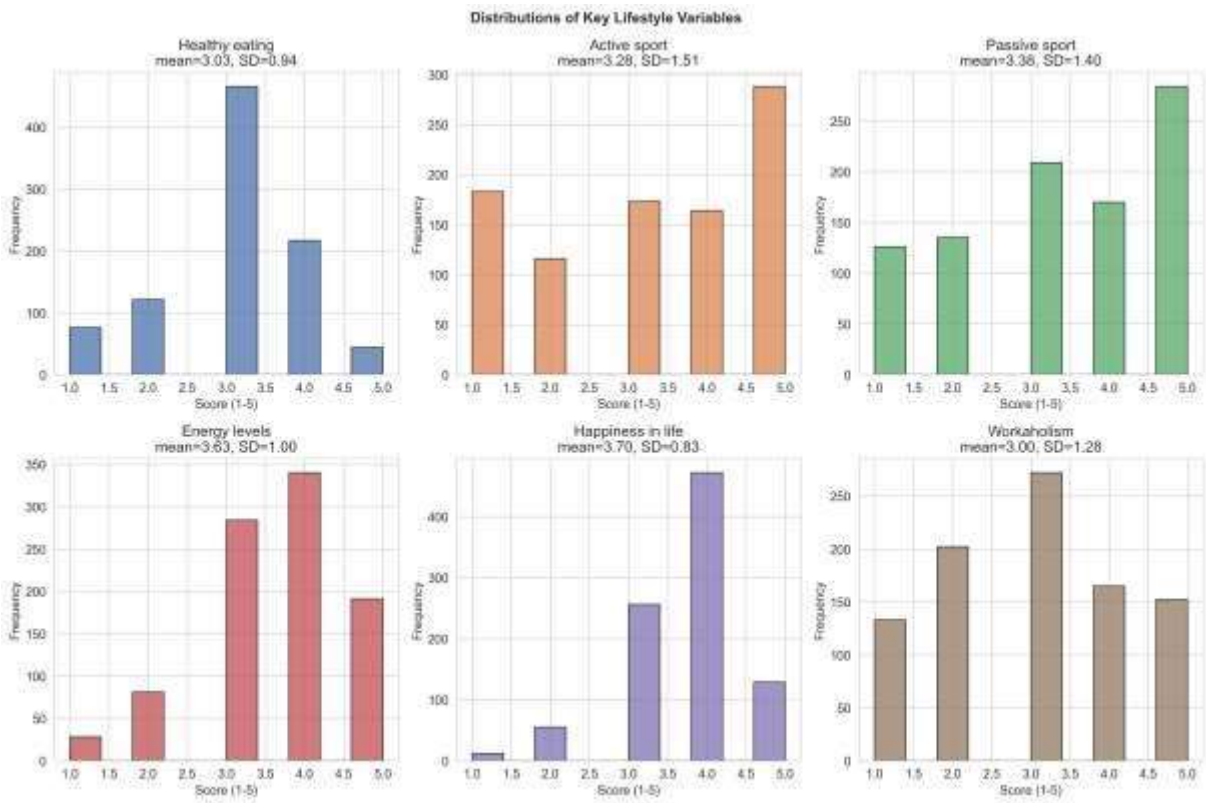


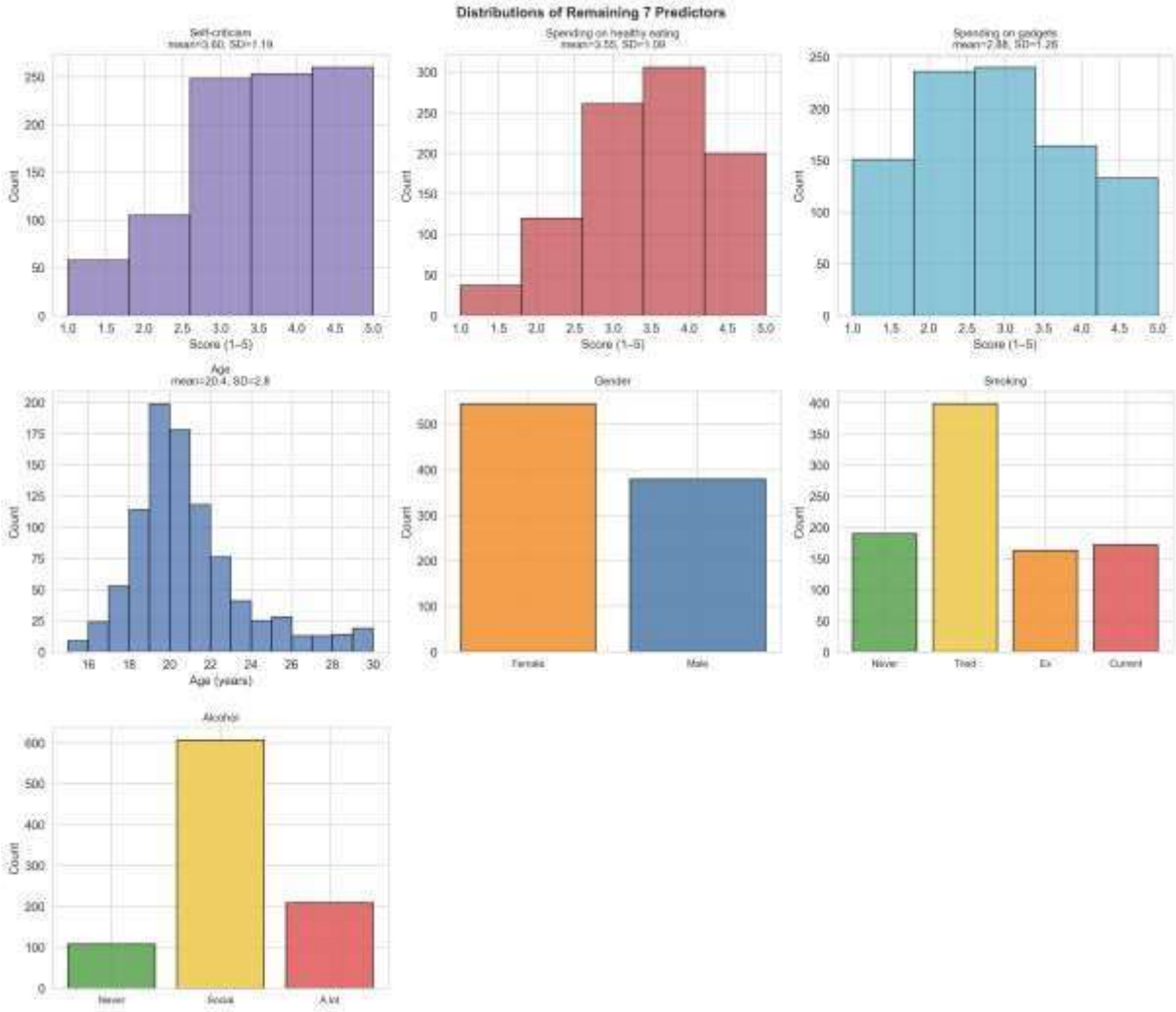
Figure 2: Distributions of six lifestyle predictors (Likert scales 1–5).

as seen in Figure 3 the remaining seven predictors, covering the three continuous or ordinal variables (Spending on healthy eating, Spending on gadgets, Self-criticism) and the remaining three categorical coded variables (Alcohol, Gender, Smoking). The mean of the Self-criticism is 3.60 (SD 1.19), so respondents are moderately self-critical on average. Spending on healthy eating is marginally above the midpoint at mean 3.55 (SD 1.09), and the spending on gadgets is below at the mean of 2.88 (SD 1.29). while spending on gadgets is below at mean 2.88 (SD 1.29). For the categorical variables: most of the respondents report that they never smoked at all or only tried one time in life; the combination of former and current smokers together is in minority. For taking Alcohol, the majority are the social drinkers, there are few they consume heavily. The gender split is 59% are females and 41% are males after I applied the BMI filters. So, this is the distributons match what would be expected from a student-aged sample.

3.2 Correlations

as seen in Figure 4 the paired Spearman correlations. The strongest correlation with BMI is Gender at r = 0.446. it is not surprising because naturally males and females are different in the body composition. The age correlates with BMI at r = 0.191. Even in between the narrow age range from 15 to 30 years, older individuals have slightly higher BMI then younger.

Spending on gadgets is r =0.179, possibly because it indirectly captures sedentary behavior. Active sport gives r = − 0.111, which is negative as expected but it is weaker than what I thought it would be.



as seen in Figure 3 Distributions of the remaining seven predictors are not shown in Figure 2: three continuous/ordinal Likert-scale variables are on(top) and three categorical variables displayed as bar charts (bottom). It is very counterintuitive. The one reason for this should be bias social desirability: peoples report they are eating healthy but practically the don't do that. So, it adds noise and it deduce the true correlation. The other possibility for this case is the scale in caloric intake because in this survey the it doesn't capture too much details about it.

When we see the correlations between the predictors themselves, Active and Passive sport shows moderate positive correlation. Alcohol and Smoking are also correlate positvely. These patterns suggest some moderate multicollinearity in the data, which is one motivation for testing Ridge regression.

3.3 Gender Differences

There are few significant differences between males and females. Female mean BMI is 20.80 and male mean is 23.32 (Mann-Whitney U, p < 0 .001). in both the Male score is higher.

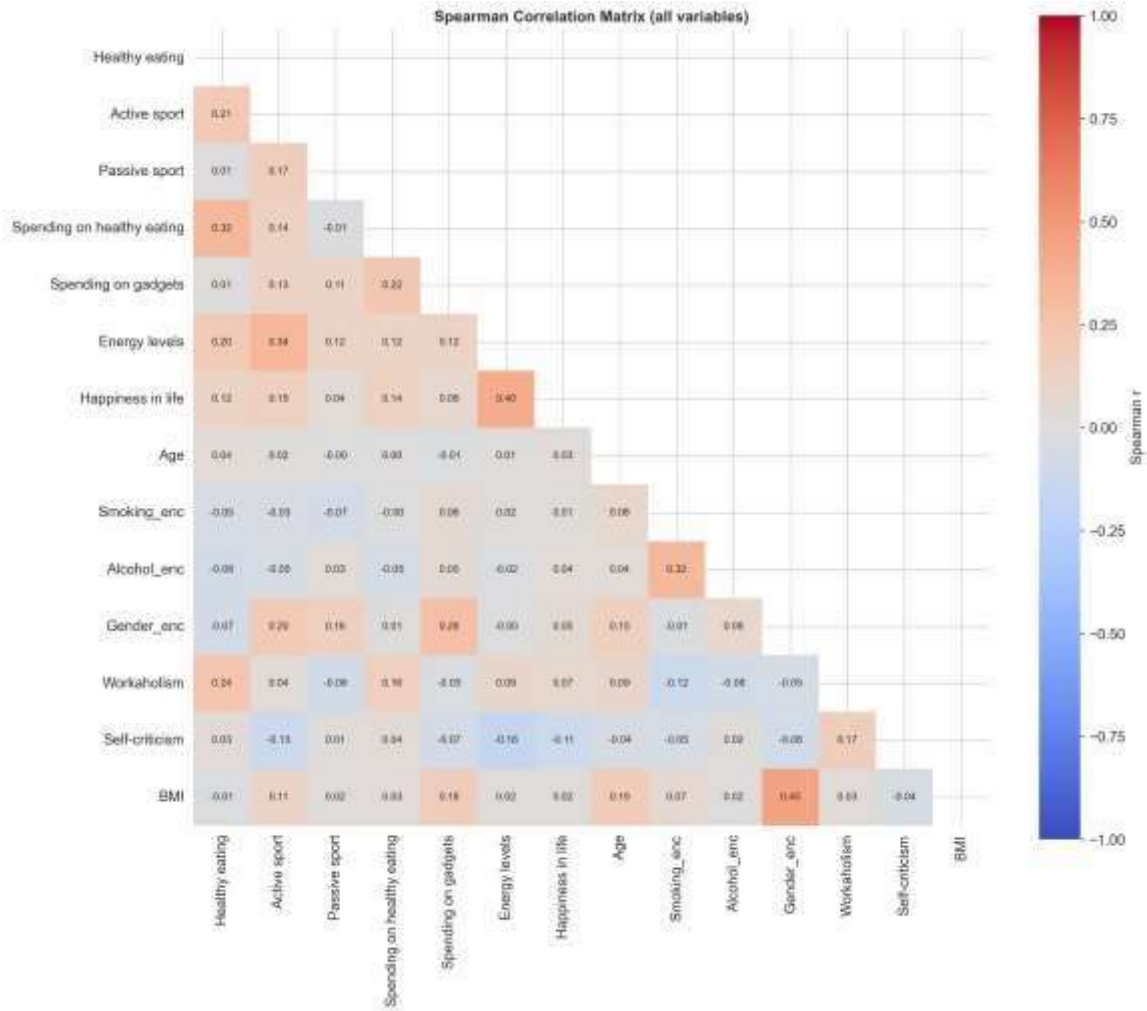


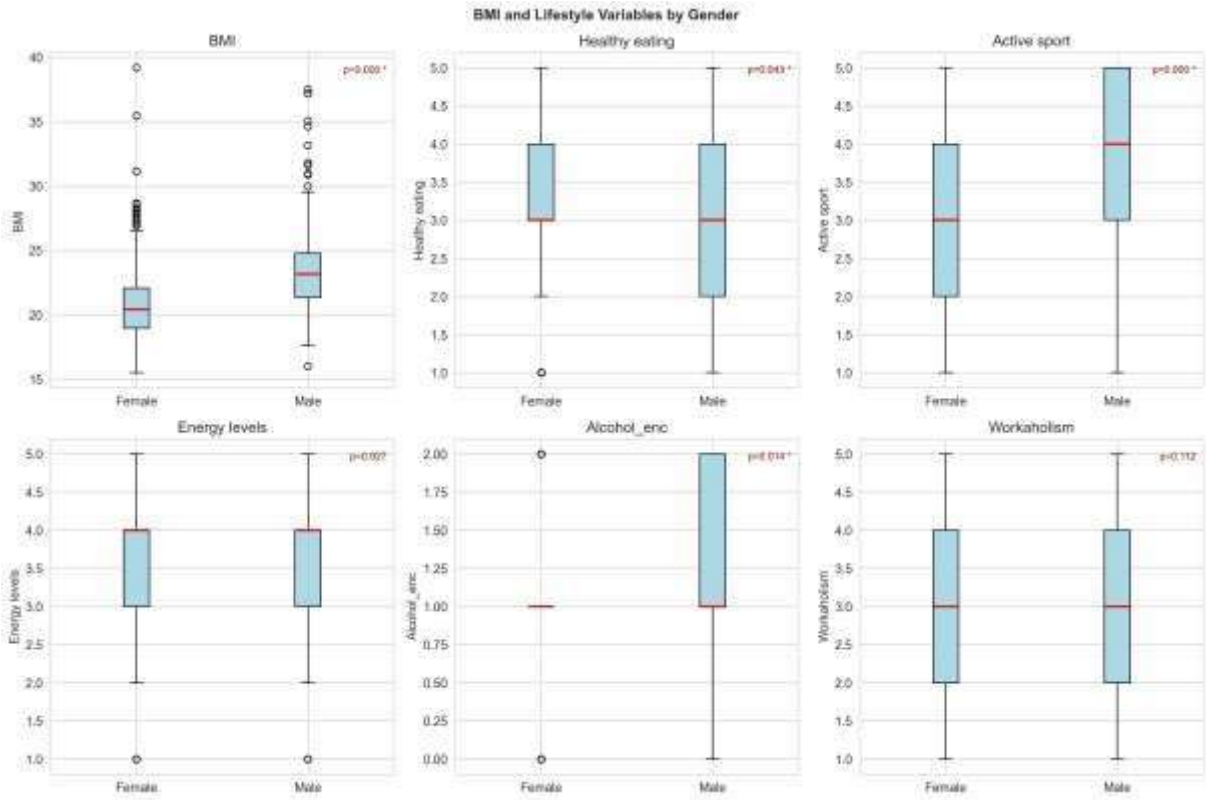
Figure 4: Spearman correlation matrix is for all 13 predictor variables and also the BMI. Only the lower triangle is shown.

Active sport (3.63 vs 3.04, p < 0 .001) and passive sport (3.64 vs 3.20, p < 0 .001). Females are slightly higher in the healthy eating (3.09 vs 2.96, p = 0.043). Alcohol consumption is also different here the males drink a bit more on average (1.16 vs 1.07, p = 0.014). Variables that do not differ by gender include Energy levels (p = 0.927), Smoking (p = 0.755), and Workaholism (p = 0.111).

The gender pattern is relevant for regression because males have higher BMI but also more physical activity. This creates a tricky situation: sport activity and BMI both differ by gender, so in multivariate analysis the gender variable will “absorb” some of the variance that might otherwise be attributed to sport.

3.4 Scatterplots

in figure 6 we see the visual of individual relationships. the gander panel shows two groups, the male group is higher and with increasing age steadily for Alcohol,



in figure 5 Figure 5: Comparison of BMI and selected lifestyle variables between females (n = 545) and males (n = 379). Asterisks show significant differences (Mann-Whitney U test, p < 0 .05).

when we look at the healthy eating and energy levels it seems very scattered plot. Abd the spearman r values are very small. When we see active sport, it is slightly downward trend but he scatters is wide. So these plots indicating that the regression model is not achieved the very high r2.

4 regression Methods

4.1 Ordinary Least Squares

Ordinary Least Squares or OLS regression finds the set of numbers they called coefficients and they are minimzed the total squared error between the predicted and actual values. So the formula is:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}\right)^2 \qquad (1)$$

I fitted the odel using the stats model library in python. The stats model library is providing p-values and 95 percent confidence intervals for each of the coefficient, which is very useful for interpretation. Then i standardized the all predictors to have a mean of 0 and the standard deviation of 1 before fitting the model. This means the coefficients of the model can be compared directly. A coefficient of 0.5 indicates an effect, than a coefficient of 0.2. they called standardized regression coefficient and sometime they aslo called beta weights.

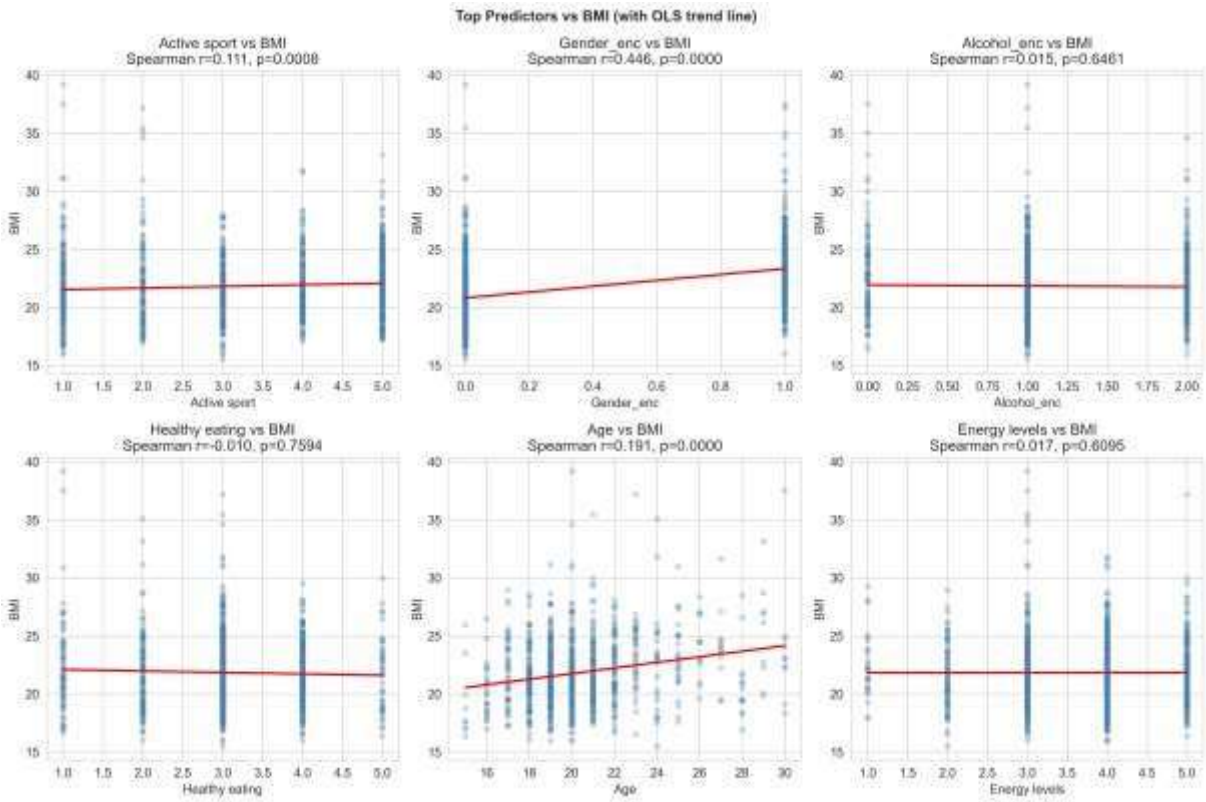


Figure 6: Scatterplots of six predictors against BMI with trend lines. Spearman r and p-values are shown in each plot title.

4.2 Ridge Regression

Ridge regression add a regularisation penalty to the OLS loss:

$$\hat{\beta}_{\text{Ridge}} = \underset{\beta}{\operatorname{argmin}} \left[\sum_{i=1}^{n} \left(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}\right)^2 + \alpha \sum_{j=1}^{p} \beta_j^2 \right] \qquad (2)$$

The term $\alpha \mathrm{P} \beta_j^2$ shrink all coefcients towards zero. Larger α means stronger shrinkage. The purpose is to reduce overfitting and to make coefficients more stable when predictors correlate with each other. With pure OLS produce coefficients with high variance. So they might look

very different across samples, from the same population. Here Ridge regression reduces this problem.

I am applying RidgeCV class of scikit-learn which automatically selects the best a of a grid of. candidates: {0.01,0.1,0.5,1,5,10,50,100,500}. This is selected using 10-fold cross. validation and the final performance of the two is also assessed by the same 10-fold process. models.

5 Results

5.1 OLS Regression

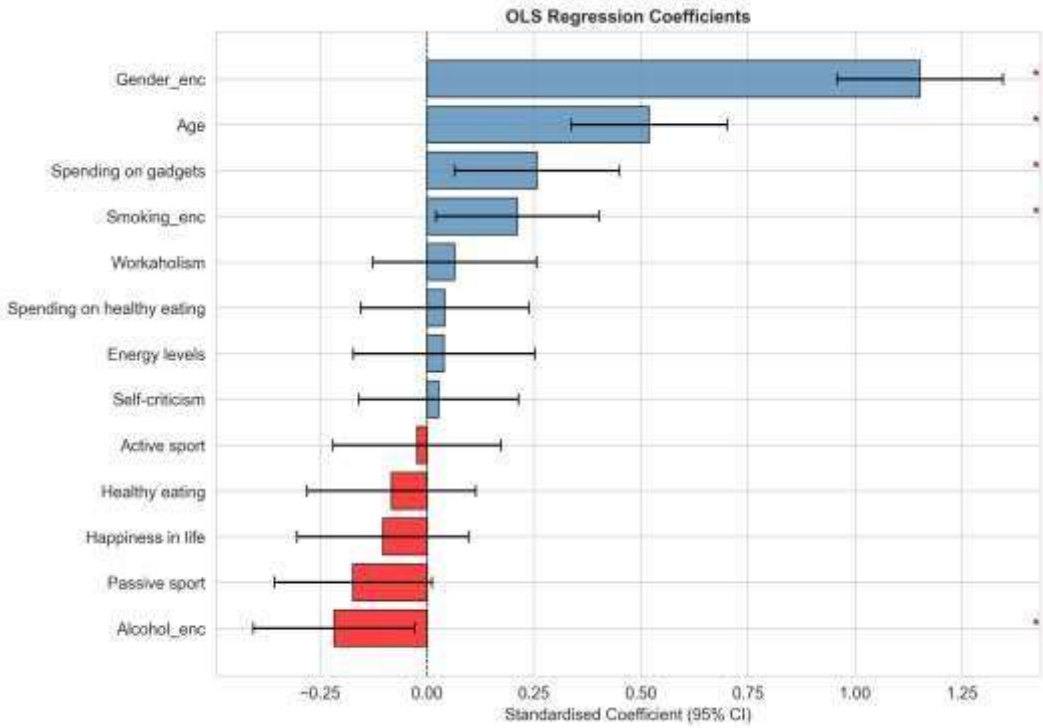


Figure 7: Standardized OLS regression coefficients with 95% confidence intervals. Blue bars are positive coefficients, red are negative. Stars show significant predictors at p < 0.05.

The OLS model explains $R^2 = 0.210$ of the variances in BMI, with adjusted $R^2 = 0.199$. The overall model is highly significant: $F\ (13,910) = 18.60$, $p < 0.001$. AIC is 4505.87 and BIC is 4573.47. In-sample prediction error is RMSE = 2.73 and MAE = 2.04 BMI units.

To be honest, 21% explained variance is not a great result. The model explains only about one fifth of why BMI differs between respondents. But I was not expecting more, because most predictors are coarse self-report scales and there are many determinants of BMI that are not measured here at all.

Table 1 shows the standardized coefficients for all 13 predictors. Five of them reach statistical significance at $p < 0.05$.

Gender is clearly the strongest predictor (coef = 1.152, p < 0.001). Being male is associated with about 2.5 extra BMI points on average after controlling for all other variables. This is by far the biggest effect in the model.

Age is the second strongest predictor (coef = 0.519, p < 0.001). Within this 15–30-year group, each standard deviation increase in age corresponds to 0.52 standard deviations more BMI. On the raw scale this is roughly one extra BMI point per few years.

Spending on gadgets is a significant positive predictor (coef = 0.258, p = 0.009). I read this Table 1: OLS regression results, standardized coefficients (n = 924). Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Predictor	Coef.	CI low	CI high	
Gender (male = 1)	1.152	0.958	1.346	***
Age	0.519	0.337	0.701	***
Spending on gadgets	0.258	0.065	0.451	**
Smoking	0.212	0.021	0.402	*
Alcohol	-0.217	-0.405	-0.028	*
Passive sport	-0.173	-0.357	0.012	
Happiness in life	-0.103	-0.303	0.098	
Healthy eating	-0.083	-0.280	0.114	
Workaholism	0.065	-0.126	0.257	
Energy levels	0.040	-0.172	0.252	
Spending on healthy eating	0.042	-0.154	0.238	
Self-criticism	0.028	-0.159	0.214	
Active sport	-0.023	-0.219	0.173	

as indirect evidence that sedentary screen-based behavior is related to higher BMI. This is perhaps the most “lifestyle” predictor in the significant group.

Smoking shows a positive coefficient (coef = 0.212, p = 0.029). This direction was surprising for me. In many published studies smoking is associated with lower body weight. But in this young Slovak sample the direction is reversed. Maybe it reflects different demographics here, or maybe heavier individuals are for some reason more likely to smoke in this sample. It is important to note that this is observational data and we cannot make causal claims.

Alcohol has a significant negative coefficient (coef = −0.217, p = 0.025). Higher alcohol use is associated with slightly lower BMI after we control for other variables. This too is

counterintuitive. One possible explanation is that in this young sample, alcohol use is correlated with being more socially and physically active, which could offset any caloric effect. Another explanation is calorie substitution people who drink more may eat less. We simply cannot know from this data which is true.

The eight non-significant predictors include Healthy eating (coef = −0.083, p = 0.409), Active sport (−0.023, p = 0.818), Passive sport (−0.173, p = 0.067), and all personality variables. The non-result for Active sport is perhaps the most interesting finding. In the bivariate correlation it had r = −0.111 with BMI, but in the multivariate model its coefficient is essentially zero. This suggest that the bivariate relationship between sport and BMI is largely mediated by gender: males exercise more and also have higher BMI, so when gender is in the model, sport no longer adds independent information.

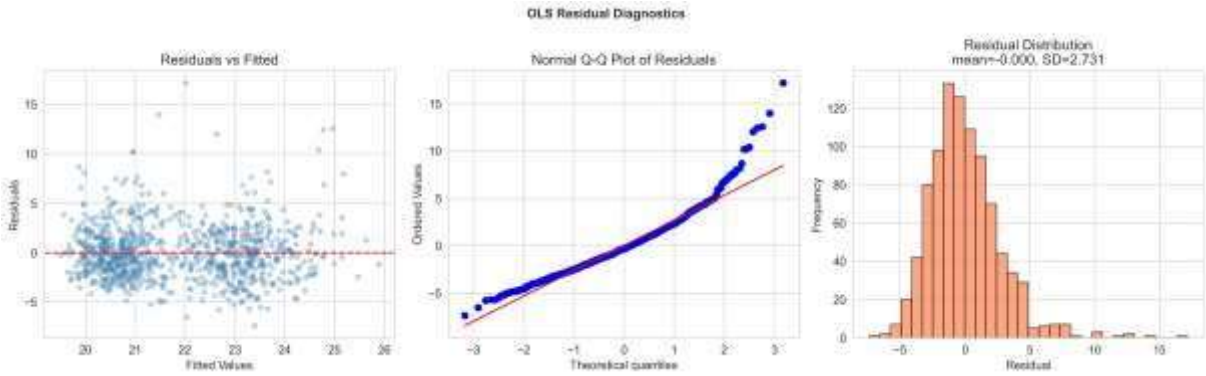


Figure 8: OLS residual diagnostics. Left: residuals vs fitted values. Middle: normal Q-Q plot of residuals. Right: residual frequency histogram.

5.2 Residual Diagnostics

The Q-Q plot and histogram show that residuals are not normally distributed. They have a right tail that is heavier than normal. The Shapiro-Wilk test confirms this: W = 0.9604, p < 0.001. The Durbin-Watson statistic is 1.96, which is very close to 2 and means there is no meaningful autocorrelation in the residuals.

To test the homoscedasticity assumption formally I ran a Breusch-Pagan test. The result is LM statistic = 47.73, p < 0.001, which means the assumption of constant residual variance is rejected. This is not entirely surprising: the skewed BMI distribution means the model underpredicts extreme values and those large residuals cluster at the upper end of the fittedvalue range.

To assess the practical impact, I compared conventional OLS standard errors with HC3 heteroscedasticity robust standard errors for the five significant predictors. The robust SEs are 1–16% larger than the conventional ones (Gender +0.9%, Age +15.9%, Spending on gadgets +9.6%, Smoking +10.7%, Alcohol +9.4%). All five predictors remain statistically significant even with the larger robust

standard errors, so the heteroscedasticity does not overturn any main conclusion. Still, it should be noted that the reported p-values and confidence intervals in Table 1 are slightly optimistic, and a fully rigorous analysis would report the HC3-based intervals instead.

5.3 Ridge Regression

The cross-validated Ridge model selects $\alpha = 100.0$ as the best regularization strength. This is quite a high value. It means the predictors have some multicollinearity and the OLS solution benefit from shrinkage.

In-sample performance is $R^2 = 0.208$ and RMSE = 2.73, nearly identical to OLS. The coefficient pattern in Ridge is very similar to OLS. Gender (Ridge coef = 1.031) and Age (0.483) are still the two dominant predictors. Spending on gadgets (0.258), Smoking (0.183), and Alcohol (−0.179) form the next group. The weak predictors like Active sport (0.000!), Self-criticism (0.016), and Energy levels (0.019) are shrunk almost all the way to zero. This actually validate the OLS result that these variables add very little Ridge confirms they can be essentially ignored without loss of prediction.

6 Model Comparison

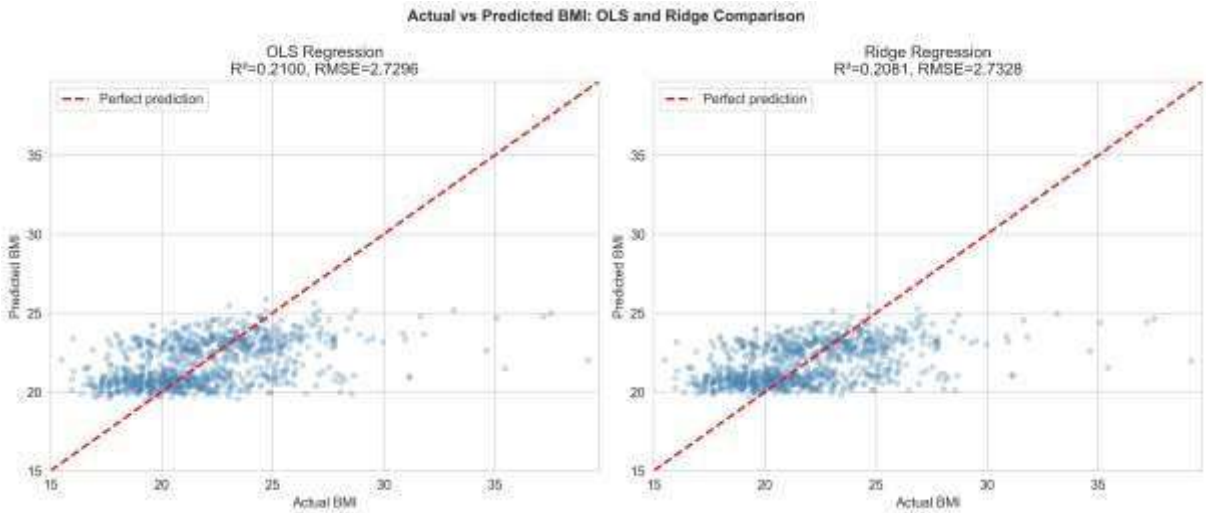


Figure 9: Actual vs predicted BMI for OLS (left) and Ridge (right). The dashed diagonal is the line of perfect prediction. Both models produce a similar scatter pattern.

Table 2 shows the 10-fold cross-validation results for both models.

Table 2: 10-fold cross-validated performance of OLS and Ridge (mean and SD over 10 folds).

Metric	OLS mean	OLS SD	Ridge mean	Ridge SD
$R2$	0.181	0.042	0.183	0.038

RMSE	2.764	0.253	2.762	0.254
MAE	2.073	0.118	2.075	0.117

The cross-validation results show that both models perform almost the same. OLS gets mean $R^2 = 0.181$ (SD = 0.042) and Ridge gets $R^2 = 0.183$ (SD = 0.038). The RMSE values are 2.764 for OLS and 2.762 for Ridge, which is not a meaningful difference. MAE is also nearly identical.

To test whether this difference is larger than sampling noise I ran a paired t-test on the 10 perfold R^2 values. The result is $t = 0.923$, $p = 0.380$, meaning the average advantage of Ridge ($+0.0017$ per fold) is not statistically significant at any conventional level. So, the mean R^2 difference should not be claimed as evidence that Ridge is genuinely better at predicting.

However, looking at the standard deviations, Ridge is noticeably more consistent across folds. Its R^2 SD (0.038) is 9.7% smaller than OLS (0.042). In other words, on a fold-to-fold basis Ridge varies less, even though the average is similar. In the actual vs predicted plots (Figure 9) both models look very similar predictions concentrate around BMI 20-25 and both struggles to capture the higher values above 30.

On the basis of this I select Ridge as the preferred model for *deployment*, not because it is measurably more accurate but for two practical reasons. First, the fold-to-fold variance is lower (SD 0.038 vs 0.042), which means Ridge predictions would fluctuate less when applied to new samples. Second, and more compelling, the optimal regularization strength is $\alpha = 100$ a high value that was chosen by cross-validation from a grid spanning four orders of magnitude. A model that benefits most from heavy shrinkage is indicating that the raw OLS coefficients are somewhat unstable due to predictor correlations. The cross-correlation evidence from the EDA (Alcohol–Smoking $r = 0.28$, Active sport–Energy levels $r = 0.34$, Spending variables intercorrelated) is consistent with this.

That said, for the purpose of interpretation I use the OLS model because it provides p-values and confidence intervals for individual coefficients. Ridge coefficients cannot be tested for significance in the standard way. So, the OLS Table 1 remains the right tool for answering which specific predictors are important, while Ridge would be the right choice if the goal were purely predictive deployment.

7 Limitations

I want to be honest about several problems with this analysis.

The biggest issue is measurement. Almost all predictor variables are self-reported on 1–5 Likert scales. Self-report data is known to be noisy and often biased. People tend to say they eat healthier food and do more exercise than they actually do. This kind of measurement error

push coefficients towards zero the technical term is attenuation bias. So, the actual relationships between lifestyle habits and BMI are probably stronger than what we see in this model. This might explain why Active sport and Healthy eating are not significant: not because they have no real effect on body weight, but because the way they are measured here is too imprecise.

Related to this: the “spending on gadgets” variable is not a direct measure of sedentary behavior. It captures how much money people spend on electronics. Some respondents might spend a lot on gadgets because they are genuinely very sedentary, but others might just have a higher income or like technology as a hobby. So, the interpretation of this predictor is uncertain.

The sample is also limited. It consists of young adults from Slovakia, mostly students, with an age range of 15 to 30. These results may not apply to older adults, people with different educational backgrounds, or people from other countries. In an older sample, for example, the relationship between lifestyle and BMI might be much stronger because habits accumulate over more years.

The R^2 of 0.21 also remind us that genetics, metabolic rate, and medical conditions are probably much more important determinants of BMI than any behavioral variable in this survey. We are not measuring those things.

A technical limitation: I treat Smoking and Alcohol as continuous ordinal variables by assigning them numeric codes. This assume that the distance between “never smoked” and “tried smoking” is the same as between “tried smoking” and “former smoker”, which may not be true. Using separate dummy variables for each category would be more flexible, but would also add complexity to the model.

A technical limitation worth naming directly is the violation of two OLS assumptions. The Breusch-Pagan test gives LM = 47.73 ($p < 0.001$), confirming significant heteroscedasticity. The residuals are also non-normally distributed (Shapiro-Wilk W = 0.9604, $p < 0.001$). Both issues make the reported standard errors and confidence intervals slightly too narrow. I quantified the effect using HC3 robust standard errors: for the five significant predictors the conventional SEs are 1–16% smaller than the robust ones, and all five predictors remain significant either way. So, the conclusions are not changed by this violation, but a fully rigorous analysis should report HC3-based intervals.

8 Conclusion

I used OLS regression and Ridge regression to see whether lifestyle habits can predict BMI in young people. The dataset came from the Young People Survey, and after removing unrealistic BMI values and rows with missing data the final sample was $n = 924$.

Both models gave quite similar results in 10-fold cross-validation. Ridge had a slightly better mean R^2 (0.183, SD = 0.038) compared to OLS (0.181, SD = 0.042), and the fold-to-fold variation was a bit smaller. Based on this I picked Ridge as the better model, also because the selected $\alpha = 100$ shows that regularisation is actually doing something useful. The prediction error for both was around RMSE = 2.76 BMI units.

For interpretation I mainly look at the OLS result because it gives p-values and confidence intervals. Gender and age dominate the model completely. Being male is associated with about 2.5 more BMI points (coef = 1.152, $p < 0.001$, CI [0.958, 1.346]). Age also has a strong positive effect (coef = 0.519, $p < 0.001$). Spending on gadgets, Smoking, and Alcohol also reach significance, but their coefficients are much smaller.

For me the most interesting part of the result is what is not significant. Healthy eating ($p = 0.409$) and Active sport ($p = 0.818$) both fail to predict BMI once gender and age are in the model. This is not what most people would guess. I think the reason is mainly measurement a single 1 5 question about how healthy someone eats simply cannot capture the actual variety of diets across people. The same goes for sport.

Overall, the 21% explained variance is modest. For a study using self-reported behavioral data this is not unusual. Genetics, metabolic rate and other unmeasured factors probably explain a lot more of the variation in BMI than any behavior survey can.

Returning directly to the research question *Which self-reported lifestyle habits are associated with BMI, and how well can we predict BMI from these habits?* the answer the data gives is largely negative for the behavioral variables. Once gender and age are in the model, only Spending on gadgets (a proxy for sedentary behavior), Smoking, and Alcohol remain statistically significant, each with small standardised coefficients ($|\beta| < 0.26$). The variables that most people would expect to matter most Healthy eating ($p = 0.409$) and Active sport ($p = 0.818$) are not significantly associated with BMI in this multivariate model. The strongest predictors are demographic (gender, age), not behavioral. This does not mean that diet and exercise do not affect weight; rather, it means that the single-item survey questions used here are too coarse to capture those effects reliably. The honest answer to the research question is: demographic factors account for most of the explainable variance, and the lifestyle variables measured in this survey offer only modest and unreliable additional signal.